

Lecture 6

Unsupervised Learning: Clustering and PCA

Machine Learning I

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Outline

k -Means

Hierarchical Clustering

Principal Component Analysis

Bridge to Machine Learning II

No Labels

Five lectures on $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}$. Today $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$, $\mathbf{x}_i \in \mathbb{R}^p$, and nothing else.

Without a y there is no loss $L(y, \hat{y})$ and no test error. What replaces them is an **objective defined on the inputs alone**, and the two classical ones are:

Clustering

Partition the points into K groups so that points in a group are close and points in different groups are far. Objective: within-group scatter.

Dimensionality reduction

Map $\mathbb{R}^p \rightarrow \mathbb{R}^k$, $k \ll p$, losing as little as possible. Objective: reconstruction error, or retained variance.

Both are optimization problems with exact statements and, for PCA, an exact solution. Neither has a ground truth to compare against, so “as little as possible” must be defined before anything is computed.

k -Means

The Objective

Assign each point to one of K clusters, $c(i) \in \{1, \dots, K\}$, with centroids $\mu_1, \dots, \mu_K \in \mathbb{R}^p$.

Within-cluster sum of squares

$$J(c, \mu) = \sum_{i=1}^N \left\| \mathbf{x}_i - \mu_{c(i)} \right\|^2 = \sum_{k=1}^K \sum_{i: c(i)=k} \left\| \mathbf{x}_i - \mu_k \right\|^2.$$

- Minimizing over c and μ jointly is NP-hard, even for $K = 2$ in the plane (Aloise et al., 2009).
- But for either variable *with the other fixed* it is trivial. That is the algorithm.

The Two Half-Steps

Fix μ , minimize over c

The sum separates over i ; the i -th term depends only on $c(i)$:

$$c(i) = \arg \min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2.$$

Assign every point to its nearest centroid.

Fix c , minimize over μ

The sum separates over k . For cluster k with N_k members,

$$\nabla_{\boldsymbol{\mu}_k} \sum_{i: c(i)=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 = -2 \sum_{i: c(i)=k} (\mathbf{x}_i - \boldsymbol{\mu}_k) = \mathbf{0} \implies \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i: c(i)=k} \mathbf{x}_i.$$

Each centroid is the mean of its cluster — hence the name.

Alternate the two until nothing changes. This is **coordinate descent** on J (Lloyd, 1957).

Numerical Example

Six points in $\mathbb{R}^2$: $A(1, 1)$, $B(1.5, 2)$, $C(3, 3.5)$, $D(5, 7)$, $E(3.5, 5)$, $F(4.5, 5)$. Initial centroids $\mu_1 = A$, $\mu_2 = D$.

	A	B	C	D	E	F	
$\ \mathbf{x} - \mu_1\ ^2$, $\mu_1 = (1, 1)$	0	1.25	10.25	52	22.25	28.25	
$\ \mathbf{x} - \mu_2\ ^2$, $\mu_2 = (5, 7)$	52	37.25	16.25	0	6.25	4.25	
assignment	1	1	1	2	2	2	$J = 11.5 + 10.5 = 22$
update: $\mu_1 = \frac{1}{3}(A + B + C) = (1.833, 2.167)$, $\mu_2 = \frac{1}{3}(D + E + F) = (4.333, 5.667)$							
$\ \mathbf{x} - \mu_1\ ^2$	2.06	0.14	3.14	33.4	10.8	15.1	
$\ \mathbf{x} - \mu_2\ ^2$	32.9	21.5	6.47	2.22	1.14	0.47	
assignment	1	1	1	2	2	2	$J = 5.33 + 3.83 = 9.17$

No assignment changed in the second pass, so the centroids will not change either: converged, $J = 9.17$.

k-Means Converges

Lloyd's algorithm terminates at a local minimum of J in finitely many steps

Proof

- Each half-step minimizes J over one block of variables with the other fixed, so J never increases: $J(c^{(t+1)}, \mu^{(t)}) \leq J(c^{(t)}, \mu^{(t)})$ and $J(c^{(t+1)}, \mu^{(t+1)}) \leq J(c^{(t+1)}, \mu^{(t)})$.
- $J \geq 0$, so the non-increasing sequence $J^{(t)}$ converges.
- There are finitely many assignments c (at most K^N), and each assignment determines its optimal μ . A strictly decreasing J cannot revisit an assignment. So after finitely many steps no assignment changes, and the algorithm halts. □

“Local minimum” is the operative phrase. Different initializations reach different partitions with different J . Run it several times and keep the best; or initialize with care (next slide).

Initialization and Choosing K

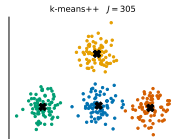
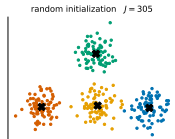
k-means++ (Arthur & Vassilvitskii, 2007)

1. Pick μ_1 uniformly from the data.
2. Pick $\mu_{j+1} = \mathbf{x}_i$ with probability $\propto \min_{l \leq j} \|\mathbf{x}_i - \mu_l\|^2$.
3. Run Lloyd's algorithm.

Far-apart seeds. Guarantee: $\mathbb{E}[J] \leq 8(\ln K + 2) J^*$ before Lloyd even starts.

Choosing K

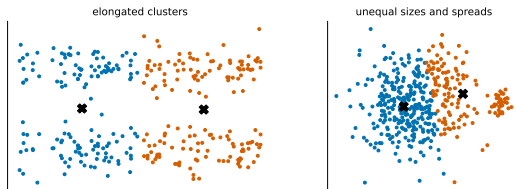
- J decreases in K always ($K = N$ gives $J = 0$). Look for the *elbow* where the decrease slows.
- Silhouette: for each point, $(b - a) / \max(a, b)$ with a its mean distance within its cluster and b to the nearest other cluster. Average over points; pick K maximizing it.
- Neither is a test error. Both are heuristics for a question the data cannot answer alone.



What k -Means Assumes

k -means is the special case of fitting a mixture of K Gaussians with identical spherical covariance $\sigma^2 \mathbf{I}$ and $\sigma \rightarrow 0$ — each half-step is a hard-assignment version of the EM algorithm. So it inherits the assumptions:

- **Spherical clusters:** Euclidean distance, so elongated clusters get cut across the long axis.
- **Similar sizes:** a large cluster's points near a small cluster get stolen by it.
- **Every point belongs somewhere:** no notion of noise or outliers; one far point drags a centroid.
- **Scale:** change the units of one feature and every distance changes. Standardize first.



Hierarchical Clustering

Agglomerative Clustering

No K up front. Start with N singleton clusters; repeatedly merge the two closest; stop when one remains. The record of merges is a **dendrogram**; cutting it at any height gives a partition.

What “closest” means: the linkage $d(A, B)$ between clusters

Single	$\min_{\mathbf{a} \in A, \mathbf{b} \in B} \ \mathbf{a} - \mathbf{b}\ $ — chains through nearby points; finds elongated shapes
Complete	$\max_{\mathbf{a} \in A, \mathbf{b} \in B} \ \mathbf{a} - \mathbf{b}\ $ — compact, roughly equal-diameter clusters
Average	$\frac{1}{ A B } \sum_{\mathbf{a}, \mathbf{b}} \ \mathbf{a} - \mathbf{b}\ $ — a compromise
Ward	the increase in within-cluster sum of squares caused by merging A and B

Cost $O(N^2 \log N)$ in time and $O(N^2)$ in memory for the distance matrix — fine to $N \sim 10^4$, not beyond.

Ward's Criterion, Derived

Ward's merge cost

Merging clusters A and B (sizes n_A, n_B , means μ_A, μ_B) increases J by $\frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2$.

Proof

The merged mean is $\mu = \frac{n_A \mu_A + n_B \mu_B}{n_A + n_B}$. For any set S with mean μ_S and any $\mathbf{m}$, $\sum_{i \in S} \|\mathbf{x}_i - \mathbf{m}\|^2 = \sum_{i \in S} \|\mathbf{x}_i - \mu_S\|^2 + |S| \|\mu_S - \mathbf{m}\|^2$ (expand; the cross term vanishes because deviations from the mean sum to zero). Apply it to A and B with $\mathbf{m} = \mu$:

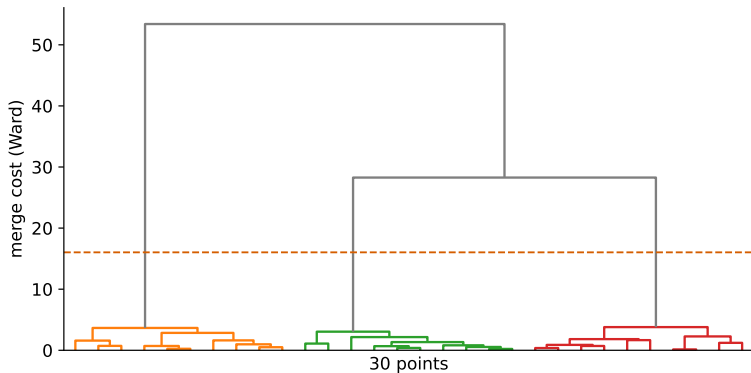
$$\Delta J = n_A \|\mu_A - \mu\|^2 + n_B \|\mu_B - \mu\|^2.$$

Now $\mu_A - \mu = \frac{n_B}{n_A + n_B} (\mu_A - \mu_B)$ and $\mu_B - \mu = \frac{n_A}{n_A + n_B} (\mu_B - \mu_A)$, so

$$\Delta J = \frac{n_A n_B^2 + n_B n_A^2}{(n_A + n_B)^2} \|\mu_A - \mu_B\|^2 = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2. \quad \square$$

Ward merges the pair whose centroids are closest, *weighted toward small clusters*: two singletons at distance d cost $d^2/2$; two clusters of 100 cost $50d^2$.

A Dendrogram



Ward linkage on 30 points from three groups. Height = merge cost. The three long vertical stretches say that three clusters is where the structure is; cutting at the dashed line recovers them.

Principal Component Analysis

The Question

Centre the data: $\bar{\mathbf{x}} = \frac{1}{N} \sum_i \mathbf{x}_i$, $\mathbf{x}_i \leftarrow \mathbf{x}_i - \bar{\mathbf{x}}$, and stack into $\mathbf{X} \in \mathbb{R}^{N \times p}$. Sample covariance:

$$\mathbf{S} = \frac{1}{N} \mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{p \times p}, \quad \text{symmetric, positive semidefinite.}$$

Two formulations of the same question

1. **Maximum variance.** Which unit direction $\mathbf{u} \in \mathbb{R}^p$ makes the projections $\mathbf{u}^\top \mathbf{x}_i$ most spread out?
2. **Minimum reconstruction error.** Which line through the origin is closest to all the points?

They have the same answer. We derive it from (1), then prove (2) is equivalent.

The First Principal Component

Maximize the projected variance

The projection of $\mathbf{x}_i$ on $\mathbf{u}$ is $z_i = \mathbf{u}^\top \mathbf{x}_i$; since the data are centred, $\bar{z} = 0$ and

$$\frac{1}{N} \sum_i z_i^2 = \frac{1}{N} \sum_i \mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} = \mathbf{u}^\top \left(\frac{1}{N} \mathbf{X}^\top \mathbf{X} \right) \mathbf{u} = \mathbf{u}^\top \mathbf{S} \mathbf{u}.$$

Maximize subject to $\|\mathbf{u}\| = 1$ (otherwise scale $\mathbf{u}$ up forever). Lagrangian and stationarity:

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{S} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1), \quad \nabla_{\mathbf{u}} \mathcal{L} = 2\mathbf{S} \mathbf{u} - 2\lambda \mathbf{u} = \mathbf{0} \implies \boxed{\mathbf{S} \mathbf{u} = \lambda \mathbf{u}}.$$

So $\mathbf{u}$ is an eigenvector of $\mathbf{S}$. At such a $\mathbf{u}$ the objective is $\mathbf{u}^\top \mathbf{S} \mathbf{u} = \lambda \mathbf{u}^\top \mathbf{u} = \lambda$. To maximize it, take the eigenvector with the **largest eigenvalue** λ_1 .

The first principal component is the top eigenvector of the covariance matrix, and λ_1 is the variance along it. Both quantities are unique whenever $\lambda_1 > \lambda_2$.

The Next Components

Write the spectral decomposition $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$, $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_p)$ orthonormal, $\lambda_1 \geq \dots \geq \lambda_p \geq 0$.

Second component: maximize variance orthogonally to $\mathbf{q}_1$

Any unit $\mathbf{u} \perp \mathbf{q}_1$ expands as $\mathbf{u} = \sum_{j \geq 2} a_j \mathbf{q}_j$ with $\sum_j a_j^2 = 1$, and

$$\mathbf{u}^\top \mathbf{S} \mathbf{u} = \sum_{j \geq 2} \lambda_j a_j^2 \leq \lambda_2 \sum_{j \geq 2} a_j^2 = \lambda_2,$$

with equality at $\mathbf{u} = \mathbf{q}_2$. By induction, the k -th component is $\mathbf{q}_k$ with variance λ_k .

All k at once (Ky Fan): $\max_{\mathbf{V}^\top \mathbf{V} = \mathbf{I}_k} \text{tr}(\mathbf{V}^\top \mathbf{S} \mathbf{V}) = \lambda_1 + \dots + \lambda_k$

Let $\mathbf{B} = \mathbf{Q}^\top \mathbf{V}$ ($p \times k$, orthonormal columns). Then $\text{tr}(\mathbf{V}^\top \mathbf{S} \mathbf{V}) = \text{tr}(\mathbf{B}^\top \mathbf{\Lambda} \mathbf{B}) = \sum_{j=1}^p \lambda_j w_j$ with $w_j = \|\mathbf{B}_j\|^2$ the squared norm of the j -th row of $\mathbf{B}$. Rows of a matrix with orthonormal columns satisfy $0 \leq w_j \leq 1$ and $\sum_j w_j = \text{tr}(\mathbf{B}^\top \mathbf{B}) = k$. A weighted sum of $\lambda_1 \geq \dots \geq \lambda_p$ with such weights is maximized by $w_1 = \dots = w_k = 1$, the rest 0: i.e. $\mathbf{V} = \mathbf{Q}_k$, the first k eigenvectors. $\square$

Variance $\Leftrightarrow$ Reconstruction

Maximizing retained variance is minimizing squared reconstruction error

Proof, for a k -dimensional subspace with orthonormal basis $\mathbf{V} \in \mathbb{R}^{p \times k}$

The projection of $\mathbf{x}_i$ onto the subspace is $\mathbf{V}\mathbf{V}^\top \mathbf{x}_i$. By Pythagoras (Lecture 1: $\mathbf{V}\mathbf{V}^\top$ is an orthogonal projection),

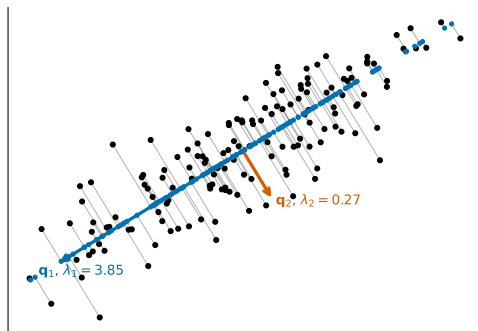
$$\left\| \mathbf{x}_i - \mathbf{V}\mathbf{V}^\top \mathbf{x}_i \right\|^2 = \left\| \mathbf{x}_i \right\|^2 - \left\| \mathbf{V}\mathbf{V}^\top \mathbf{x}_i \right\|^2 = \left\| \mathbf{x}_i \right\|^2 - \mathbf{x}_i^\top \mathbf{V}\mathbf{V}^\top \mathbf{x}_i.$$

Sum over i and divide by N :

$$\underbrace{\frac{1}{N} \sum_i \left\| \mathbf{x}_i - \mathbf{V}\mathbf{V}^\top \mathbf{x}_i \right\|^2}_{\text{reconstruction error}} = \underbrace{\frac{1}{N} \sum_i \left\| \mathbf{x}_i \right\|^2}_{\text{total variance, fixed}} - \underbrace{\text{tr}(\mathbf{V}^\top \mathbf{S} \mathbf{V})}_{\text{retained variance}}. \quad \square$$

Total variance is $\text{tr} \mathbf{S} = \sum_j \lambda_j$. Keeping k components retains $\lambda_1 + \dots + \lambda_k$ and loses $\lambda_{k+1} + \dots + \lambda_p$. The **proportion of variance explained** is $\sum_{j \leq k} \lambda_j / \sum_j \lambda_j$.

PCA in Two Dimensions



Centred data, the two eigenvectors of $\mathbf{S}$ scaled by $\sqrt{\lambda_j}$, and the projection onto $\mathbf{q}_1$. The residuals (grey) are perpendicular to $\mathbf{q}_1$ — the reconstruction-error view.

PCA via the SVD

Never form $\mathbf{S}$ and diagonalize it. Take the thin SVD of the centred data matrix:

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top, \quad \mathbf{U} \in \mathbb{R}^{N \times p}, \mathbf{D} = \text{diag}(d_1 \geq \dots \geq d_p \geq 0), \mathbf{V} \in \mathbb{R}^{p \times p}.$$

The right singular vectors are the principal components

$$\mathbf{S} = \frac{1}{N} \mathbf{X}^\top \mathbf{X} = \frac{1}{N} \mathbf{V} \mathbf{D} \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top = \mathbf{V} \frac{\mathbf{D}^2}{N} \mathbf{V}^\top,$$

which is the spectral decomposition of $\mathbf{S}$ with $\mathbf{Q} = \mathbf{V}$ and $\lambda_j = d_j^2 / N$.

- **Loadings:** columns of $\mathbf{V}$. **Scores** (coordinates of each point on the components): $\mathbf{Z} = \mathbf{X}\mathbf{V} = \mathbf{U}\mathbf{D}$.
- Cost $O(Np^2)$, numerically stable, and it gives the rank- k truncation directly (next slide).
- Lecture 3's ridge shrinks direction j by $d_j^2 / (d_j^2 + \lambda)$. **Principal components regression** keeps directions $1, \dots, k$ and drops the rest: a hard threshold where ridge is a soft one.

Eckart–Young: The Best Low-Rank Approximation

Theorem (Eckart–Young–Mirsky, 1936)

Among all matrices $\mathbf{B}$ of rank at most k , the one minimizing $\|\mathbf{X} - \mathbf{B}\|_F$ is the truncated SVD $\mathbf{X}_k = \mathbf{U}_k \mathbf{D}_k \mathbf{V}_k^\top$, and the minimum is $\|\mathbf{X} - \mathbf{X}_k\|_F^2 = \sum_{j>k} d_j^2$.

Proof for $\mathbf{B}$ of the form $\mathbf{X}\mathbf{P}$, $\mathbf{P}$ a rank- k orthogonal projection (the PCA case)

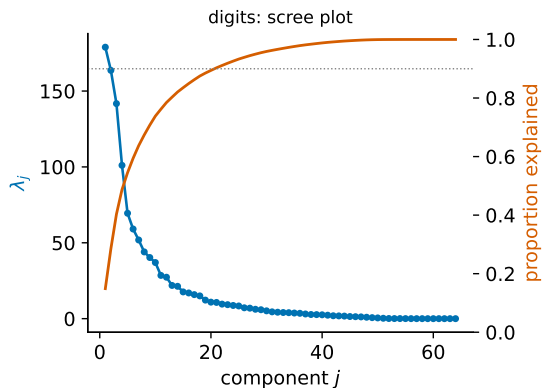
This is the previous two slides in matrix form. Write $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ with $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_k$. Then

$$\|\mathbf{X} - \mathbf{X}\mathbf{V}\mathbf{V}^\top\|_F^2 = \|\mathbf{X}\|_F^2 - \text{tr}(\mathbf{V}^\top \mathbf{X}^\top \mathbf{X} \mathbf{V}) = \sum_j d_j^2 - N \text{tr}(\mathbf{V}^\top \mathbf{S} \mathbf{V}),$$

minimized, by Ky Fan, at $\mathbf{V} = \mathbf{V}_k$, giving $\sum_j d_j^2 - \sum_{j \leq k} d_j^2 = \sum_{j>k} d_j^2$. □

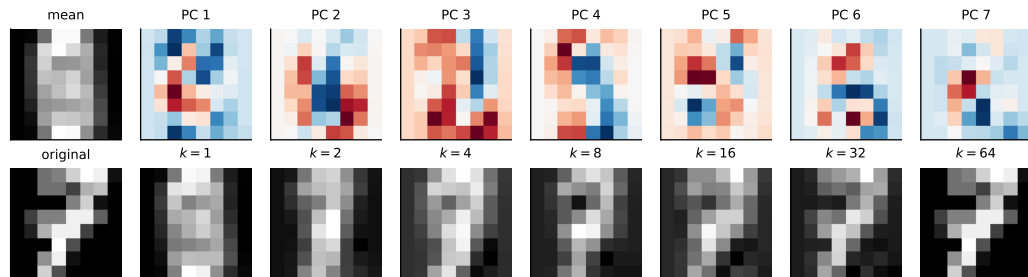
The general statement (arbitrary $\mathbf{B}$, and also the spectral norm) needs one more inequality (Weyl) and is in *ESL* §14.5. ML II uses it constantly: it is the reason low-rank factorization works.

How Many Components?



- **Scree plot:** λ_j against j . Keep components before the plot flattens.
- **Cumulative variance:** keep the smallest k that explains, say, 90%.
- **Downstream task:** if PCA feeds a supervised model, choose k by cross-validating *that* model. The only choice with a test error behind it.
- Standardize first unless the features share units: PCA chases variance, and a feature in large units has large variance for no reason.

Eigendigits



Handwritten digits, $8 \times 8 = 64$ pixels each. Top row: the mean and the first seven principal components, reshaped as images. Bottom: one digit reconstructed with $k = 1, 2, 4, 8, 16, 32, 64$ components. At $k = 16$ it is recognizable; at $k = 64$ it is exact.

Bridge to Machine Learning II

What PCA Really Is

Three readings of the same $\mathbf{X} \approx \mathbf{U}_k \mathbf{D}_k \mathbf{V}_k^\top$:

1. **Geometry**: the best k -dimensional subspace, in squared distance.
2. **Latent factors**: each observation is a combination of k hidden factors, $\mathbf{x}_i \approx \mathbf{V}_k \mathbf{z}_i$, with $\mathbf{z}_i = \mathbf{D}_k \mathbf{U}_i^\top$. The factors are unobserved; only their mixtures are.
3. **A linear autoencoder**: encode $\mathbf{z} = \mathbf{V}_k^\top \mathbf{x}$, decode $\hat{\mathbf{x}} = \mathbf{V}_k \mathbf{z}$, trained to minimize $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$.
A neural network with one linear hidden layer of width k learns exactly this subspace.

Machine Learning II starts by loosening each reading in turn: latent factors with missing entries (matrix completion), non-negative factors (NMF), then non-linear encoders and decoders — and the word “embedding” will mean a learned $\mathbf{z}_i$.

What We Proved Today

1. k -means: with centroids fixed, assign to the nearest; with assignments fixed, centroids are means. Each half-step lowers J ; finitely many assignments; so it terminates at a local minimum.
2. Ward's merge cost is $\frac{n_A n_B}{n_A + n_B} \|\boldsymbol{\mu}_A - \boldsymbol{\mu}_B\|^2$.
3. The first principal component is the top eigenvector of $\mathbf{S}$ (Lagrangian); the first k are the top k eigenvectors (Ky Fan); retained variance $= \lambda_1 + \cdots + \lambda_k$.
4. Maximum variance $\Leftrightarrow$ minimum reconstruction error (Pythagoras).
5. The right singular vectors of centred $\mathbf{X}$ are the principal components, $\lambda_j = d_j^2 / N$; the truncated SVD is the best rank- k approximation (Eckart–Young).

Next

Lab 4 (Wednesday, Rodrigo) — k -means from scratch with the convergence check, Ward's formula verified numerically, PCA by eigendecomposition and by SVD, eigendigits, and the ML I mini-project due.

Machine Learning II (Friday, October 2) — the perceptron revisited, from the other side: what happens when you stack linear models and put a nonlinearity between them.

Reading — *ESL* §14.3.6 (k -means), §14.3.12 (hierarchical), §14.5.1 (PCA), §14.5.2–14.5.3 (SVD, low rank). Arthur & Vassilvitskii (2007) for k -means++.